Lauren Jenkins

Quantitative forest ecologist | Bayesian hierarchical modeling | Global change ecology

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Education

Duke University, Nicholas School of the Environment *Durham, NC*
Ph.D. Candidate in University Program in Ecology Aug 2022 - Expected May 2027
- PI: James S. Clark
- Dissertation: *Climate-driven impacts on tree fecundity across space and time: from pollen limitation to topographic refugia*
- Certificate in College Teaching

Wheaton College *Wheaton, IL*
B.S. in Biology and Computer Science Aug 2017 – May 2021
- Math minor; graduated *cum laude*

Research Experience

Ph.D. Student *Durham, NC*
Clark Lab, Duke University Aug 2022 – present
- Developing Bayesian hierarchical models of how climate drives tree fecundity across North America and Europe, using continental-scale seed production estimates from the MASTIF network.
- Quantifying how pollen limitation constrains seed production by linking atmospheric pollen records from the National Allergy Bureau to tree-level seed estimates.
- Identifying topographic refugia where forests may persist and regenerate under future climates, by testing how terrain modifies climate effects on fecundity.
- Leading MASTIF field work trips across NEON and long-term sites to monitor tree fecundity, biodiversity, and regeneration; measuring tree growth, health, and canopy status, observational crop counts, and seedling recruitment transects.

NSF Post-Baccalaureate Biogeography Researcher *St. Louis, MO (Remote)*
Missouri Botanical Garden, Global Change and Conservation Lab Aug 2021 – Aug 2022
- Compared pollen-vegetation and species distribution models to reconstruct glacial refugia and postglacial tree migration.
- PI: Adam B. Smith

Environmental Science Consultant *Chicago, IL (Remote)*
Argonne National Laboratory, Environmental Sciences Division Sept 2020 – Aug 2023
- Analyzed ecosystem services of solar-pollinator habitats to determine insect community responses to solar energy facilities.
- Mentors: Leroy Walston, Heidi Hartmann

Research Assistant *Wheaton, IL*
Disease Ecology Lab, Wheaton College Biology Aug 2020 – May 2021
- Conducted spatial analysis of raccoon fecal samples from the Forest Preserves of Cook County to characterize parasite habitat associations.
- Mentor: Kristen Page

Blanding's Turtle Microbiome Research Assistant *Wheaton, IL*
Scott Lab, Wheaton College Biology Aug 2018 – Mar 2020
- Collected host-associated microbiome samples from endangered Blanding's turtles at Cosley Zoo and Shedd Aquarium to support genomic analysis of captivity effects.
- Mentor: Rodney Scott

Conservation Ecology Research Assistant *St. Louis, MO*
Marquis Lab, University of Missouri - St. Louis June 2019 – Aug 2019
- Designed and conducted an independent field experiment on ant seed dispersal preferences for spring-flowering woodland herbs in eastern Missouri.
- Mentor: Eva Colberg

Publications

L. Jenkins, T. Bankston, M. Hu, I. Ibáñez, R. Kays, G. Kunstler, J. Luongo, ... & J.S. Clark. "Fecundity and terrain-climate interactions define tree niche differences across North America and Europe". *Journal of Ecology*. 10.1111/1365-2745.70369 [blog post] — June 2026

J.S. Clark, T. Bankston, M. Bogdziewicz, ... **L. Jenkins** ..., & M. Żywiec. "Despite rapid warming, seed production is not leading poleward migration in North American and European forests". *New Phytologist*. 10.1111/nph.71223 — May 2026

A.J. Jensen, T. Ackley, M. Alyetama, ... **L. Jenkins** ..., & R. Kays. "A global tour of iNaturalist mammal data: status, trends, and outlook for macroecology and conservation". *Journal of Mammalogy*. 10.1093/jmammal/gyag025 — Apr 2026

J.S. Clark, R. Andrus, M. Arianoutsoue, ... **L. Jenkins** ..., & M. Żywiec. "Continental contrasts in climate extremes that control tree fecundity". *Global Change Biology*. 10.1111/gcb.70738 — Feb 2026

L. Jenkins, D.T. McKnight, M. Parks, N.W. Byer, F.J. Oliaro, D. Thompson, R. Scott. "Variable effects of captivity on microbiomes in populations of IUCN-endangered Blanding's turtles (Emydoidea blandingii)". *Journal of Applied Microbiology*. 10.1093/jambio/lxae121 — May 2024

L.J. Walston, H.M. Hartmann, L. Fox, J. Macknick, J. McCall, J. Janski, **L. Jenkins**. "If you build it, will they come? Insect community responses to habitat establishment at solar energy facilities in Minnesota, USA". *Environmental Research Letters*. 10.1088/1748-9326/ad0f72 — Dec 2023

Fellowships and Grants

- Teaching on Purpose Fellowship, Duke University: Kenan Institute for Ethics - $1,000 (2026)
- Global Student Fund, Duke Global - $500 (2026)
- Conference Travel Award, Duke University Graduate School - $525 (2026)
- Conference Travel Award, Duke University Nicholas School of the Environment - $1,500 total (2024-2026)
- Dissertation Research Domestic Travel Award, Duke University - $1,400 (2025)
- PhD Computational Summer Fellowship, Duke University Center for Computational Thinking - $10,000 (2025)
- Summer Research Fellowship, Duke University Graduate School - $22,225 total (2023, 2024)
- NSF Research Experience for Post-Baccalaureate Students - $46,076 (2021)

Awards and Honors

- NSF Graduate Research Fellowship Program Honorable Mention (2022, 2024)
- NSF Beneficial Insects Summer REU, University of Nebraska–Lincoln (awarded 2020; did not participate due to COVID-19)
- Students as Research Scientists Research Associate, University of Missouri - St. Louis (2019)

Talks and Presentations

"Climate-driven variation in tree fecundity: What controls where and when trees reproduce?" — Apr 2026
Invited talk. Wake Forest University.

"Terrain-climate interactions drive tree fecundity and tree niche differences across North America and Europe" — July 2026
Poster. Gordon Research Conference: Unifying Ecology Across Scales (Boston, MA). *Coauthor: Clark, J.S.*

"Climate refugia in North American and European temperate forests: Local habitat and terrain conditions needed for tree fecundity" — Aug 2025
Oral. Ecological Society of America Annual Meeting (Baltimore, MD). *Coauthor: Clark, J.S.*

"Mast in the Past: Changes in climate drove large shifts in seed resources since the last Ice Age" Aug 2024
Poster. Ecological Society of America Annual Meeting (Long Beach, CA). *Coauthor: Clark, J.S.*

"Mast in the Past: Predicting seed, fruit, and nut abundance since the Last Glacial Maximum using pollen-fecundity relationships" June 2024
Poster. New Phytologist Next Generation Scientists Meeting (Durham, NC). *Coauthor: Clark, J.S.*

"Species distribution models and pollen density models predict different rates of climate tracking across the last 21,000 years" Aug 2022
Oral. Ecological Society of America Annual Meeting (Montreal, QC). *Coauthors: Brown, A., Castilla A., Dawson, A., Hoban, S., Robinson, J., Strand, A., Tipton, J., Smith, A.B.*

"Dissonance in predictions from SDMs and pollen density models across the previous 21,000 years" June 2022
Oral. International Biogeography Society Meeting (Online). *Coauthors: Brown, A., Castilla A., Dawson, A., Hoban, S., Robinson, J., Strand, A., Tipton, J., Smith, A.B.*

"Ant seed preference for two spring-flowering woodland herbs" Sept 2019
Poster. Wheaton College STEM Day Summer Research Symposium (Wheaton, IL). *Coauthors: Colberg, E.M., Marquis R.J.*

"Winnow ant preference for ant-dispersed seeds, bloodroot and wild ginger" July 2019
Oral. University of Missouri Students as Research Scientists Symposium (St. Louis, MO). *Coauthors: Colberg, E.M., Marquis R.J.*

Teaching Experience

Graduate Teaching Assistant *Durham, NC*
Duke University, Nicholas School of the Environment
- **ENV 710: Applied Statistical Modeling for Environmental Management** (Fall 2022, Fall 2024, Spring 2026). Led weekly R labs for ~40 students per semester; wrote practice and exam questions; graded labs and exams.
- **ENV 102: Introduction to Environmental Sciences and Policy** (Spring 2025). Led weekly discussion sections of ~18 students; graded writing assignments and final projects.
- **BIO 341: Plant Communities of North Carolina** (Fall 2024). Supported weekly field trips with plant identification and fieldwork; wrote exam questions; graded assignments.
- **Nicholas School Statistics and Calculus Diagnostic Exams** (Summer 2024). Developed practice tests, Canvas review modules, and instructional videos used annually by ~150 incoming students; led review and help sessions.
- **Additional appointments:** ENV 646: Urban Ecology (Fall 2026); IDS 707: Data Visualization and Storytelling (Fall 2025); ENV 623: Ecological Diversity and Climate Change (Spring 2023); ENV 89S: Environmental Change in the Big Data Era (Spring 2023)

Guest Lecturer
Duke University
- Urban Vegetation (September 2026)
- Introduction to Bayesian Statistics (April 2026)
- When you can't observe what you want to know: Bayesian inference for forest seed production (March 2026)
- Community Ecology: Forests and Seed Production (November 2025)
- Paleoecology of North Carolina Plant Communities (November 2024)
- GJAM for Land and Water Geospatial Analysis (March 2023)

Teaching Training
Duke University
- Preparing Future Faculty (2026-2027)
- Teaching on Purpose Fellowship, Kenan Institute for Ethics (2026)
- Seminar in Teaching Biology (2024)

Undergraduate Lab Teaching Assistant *Wheaton, IL*
Wheaton College Biology Department Jan 2019 – May 2021
- Courses: Introduction to Botany & Zoology, Experimental Design & Research, Ecology & Evolution, Disease Ecology

Relevant Coursework and Workshops

Ecology: Functional Ecology of Plants (2022); Population Ecology (2024); Food Web Theory (2024)

Methods: Bayesian Inference for Environmental Modeling (2024); Remote Sensing for Environmental Analysis (2023); RangeshiftR Workshop, International Biogeography Society (2022); Biosphere2 Biogeography, Tucson, AZ (2023)

Science Communication: Writing and Publishing in Science (2025); ComSciCon-Triangle Workshop, Research Triangle Institute International (2024)

Equity, Inclusion, and Field Safety: Social Field Safety, Field Inclusive (2026); Preventing Sexual Harassment and Assault in the Field Sciences, Field Futures (2023); Racial Equity Learning Arc, The Equity Paradigm (2023); Decolonizing Ecology with Dr. Madhusudan Katti, Duke University (2022)

Service

University Program in Ecology Executive Committee, Student Representative 2025 – present
Duke University Program in Ecology

Henry J. Oosting Memorial Lecture, Student Organizer 2025
Duke University, Nicholas School of the Environment

Next Generation Scientists Organizing Committee, Local Organizer 2023 – 2024
New Phytologist

Ecology Seminar Planning Committee, Co-Coordinator 2022 – 2024
Duke University Program in Ecology

Recruitment Committee, Co-Coordinator 2023
Duke University Program in Ecology

Outreach

Scientific Research and Education Network (SciREN) - Triangle 2023 – 2026
Logistics Lead
- Coordinated 15-20 PhD students to plan an annual event connecting 300+ K-12 NC educators with current research for use in their classrooms, raising $7,000-10,000 in support each year.
- Authored a K-5 lesson plan on seed dispersal in trees (2023).

IGNITE Engineering Education Program, Duke University 2025 – 2026
Research Team Evaluator
- Evaluated STEM outreach for 20+ Durham-area middle schoolers.

North Carolina SciMatch for Middle School 2025
Scientist Volunteer
- Taught virtual climate change lessons to 4 classes of 8th graders.

Elementary Environmental Mentoring (ELEMENT), Durham Public Schools 2023 – 2024
Mentor
- Mentored 13 high school students through creating a lesson plan to teach elementary schoolers.

Duke-Durham SchoolDays 2023
Presenter
- Led an outdoor research experience for first-generation 8th graders.

SciComm Video Workshop, Duke University 2023
Organizer
- Awarded a $1,000 Dean's Award.

Skills

Statistical Modeling: Bayesian hierarchical models; species distribution models; pollen-vegetation models; GJAM

Programming: R, Python, C, Java

Geospatial: ArcGIS, ENVI, Google Earth Engine (JavaScript and Python)

Computing and Reproducibility: Git/GitHub; high-performance computing (cluster); R Markdown; RStudio; Jupyter Notebook; LaTeX; HTML/CSS/JavaScript